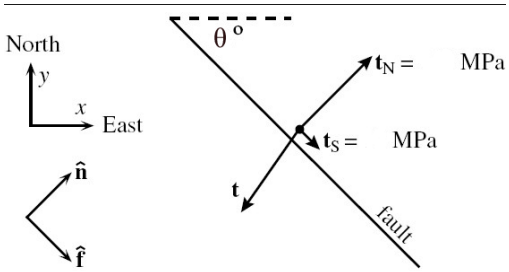
Due date (2026.3.16)

1. 已知应力张量为$\begin{pmatrix} -40 & -10 \\ -10 & -60 \end{pmatrix}$，断层的倾角θ如图所示。1）求断层面上的正应力和剪应力，用斜截面应力和坐标变换两种方法分别求取，将计算结果按照倾角 5 度为间隔编程画图展示；2）求主应力及其对应方向；3）求最大剪应力及其对应方向。



2. 题目见 Peter Shearer 书的 P35 页第 12 题，具体如下：

12. The University of California, San Diego, operates the Piñon Flat Observatory (PFO)in the mountains northeast of San Diego (near Anza). Instruments include high-quality strain meters for measuring crustal deformation.

    (a) Assume, at 5 km depth beneath PFO, the seismic velocities are $\alpha = 6$ km/s and $\beta = 3.5$ km/s and the density is $\rho = 2.7$ Mg/m^3. Compute values for the Lamé parameters, λ and μ, from these numbers. Express your answer in units of pascals.

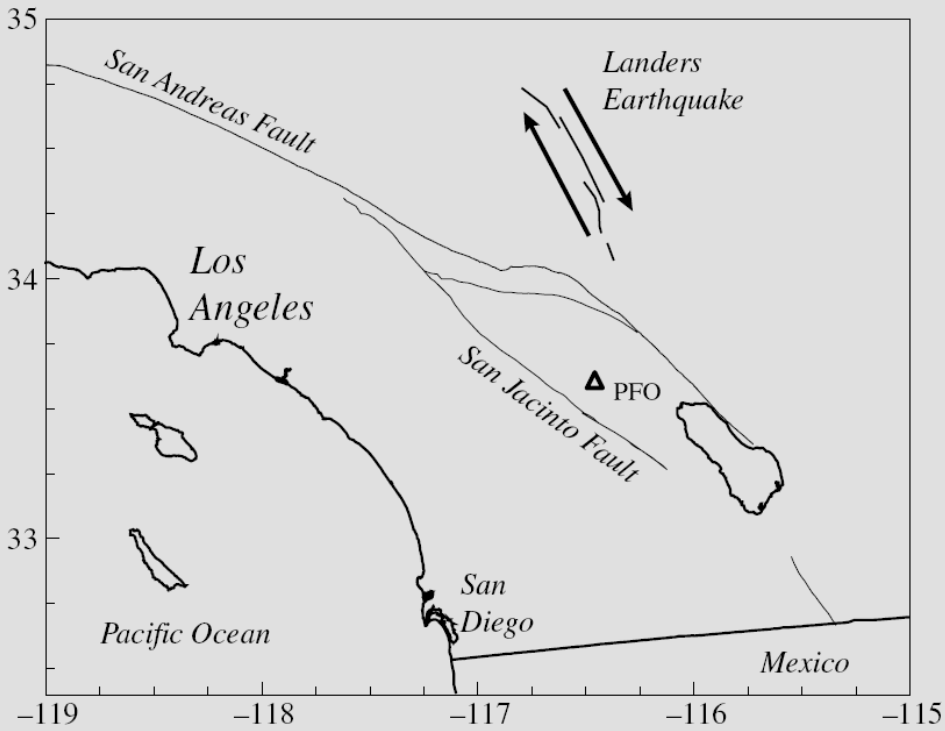


Figure 2.7 The 1992 Landers earthquake ($M_S = 7.3$) in southern California produced measurable strain changes at PFO observatory, located about 80 km south of the event.

(b) Following the 1992 Landers earthquake ($M_S = 7.3$), located in southern California 80 km north of PFO (Fig. 2.7), the PFO strain meters measured a large static change in strain compared to values before the event. Horizontal components of the strain tensor changed by the following amounts: $e_{11} = -0.26 \times 10^{-6}$, $e_{22} = 0.92 \times 10^{-6}$, $e_{12} = -0.69 \times 10^{-6}$. In this notation 1 is east, 2 is north, and extension is positive. You may assume that this strain change occurred instantaneously at the time of the event. Assuming these strain values are also accurate at depth, use the result you obtained in part (a) to determine the change in stress due to the Landers earthquake at 5 km, that is, compute the change in τ_{11}, τ_{22}, and τ_{12}. Treat this as a two-dimensional problem by assuming there is no strain in the vertical direction and no depth dependence of the strain.

(c) Compute the orientations of the principal strain axes (horizontal) for the response at PFO to the Landers event. Express your answers as azimuths (degrees east of north).

(d) A steady long-term change in strain at PFO has been observed to occur in which the changes in one year are: $e_{11} = 0.101 \times 10^{-6}$, $e_{22} = -0.02 \times 10^{-6}$, $e_{12} = 0.005 \times 10^{-6}$. Note that the long-term strain change is close to simple E–W extension. Assuming that this strain rate has occurred steadily for the

last 1000 years, from an initial state of zero stress, compute the components of the stress tensor at 5 km depth. (Note: This is probably not a very realistic assumption!) Don't include the large hydrostatic component of stress at 5 km depth.

(e) Farmer Bob owns a 1 km^2 plot of land near PFO that he has fenced and surveyed with great precision. How much land does Farmer Bob gain or lose each year? How much did he gain or lose as a result of the Landers earthquake? Express your answers in m^2.

(f) (COMPUTER) Write a computer program that computes the stress across vertical faults at azimuths between 0 and 170 degrees (east from north, at 10 degree increments). For the stress tensors that you calculated in (b) and (d), make a table that lists the fault azimuth and the corresponding shear stress and normal stress across the fault (for Landers these are the stress changes, not absolute stresses). At what azimuths are the maximum shear stresses for each case?

(g) (COMPUTER) Several studies (e.g., Stein *et al.*, 1992, 1994; Harris and Simpson, 1992; Harris *et al.*, 1995; Stein, 1999; Harris, 2002) have modeled the spatial distribution of events following large earthquakes by assuming that the likelihood of earthquake rupture along a fault is related to the *Coulomb failure function* (CFF). Ignoring the effect of pore fluid pressure, the change in CFF may be expressed as:

$$\Delta \mathrm{CFF} = \Delta|\tau_s| + \mu_s \Delta \tau_n,$$

where τ_s is the shear stress (traction), τ_n is the normal stress, and μ_s is the coefficient of static friction (don't confuse this with the shear modulus!). Note that CFF increases as the shear stress increases, and as the compressional stress on the fault is reduced (recall in our sign convention that extensional stresses are positive and compressional stresses are negative). Assume that $\mu_s = 0.2$ and modify your computer program to compute ΔCFF for each fault orientation. Make a table of the yearly change in ΔCFF due to the long-term strain change at each fault azimuth.

(h) (COMPUTER) Now assume that the faults will fail when their long-term CFF reaches some critical threshold value. The change in time to the next earthquake may be expressed as

$$\Delta t = \frac{\mathrm{CFF}_{1000+L} - \mathrm{CFF}_{1000}}{\mathrm{CFF}_a},$$

where CFF_a is the annual change in CFF, CFF_{1000} is the thousand year change in CFF, and CFF_{1000+L} is the thousand year + Landers change in CFF (note that $\mathrm{CFF}_{1000+L} \neq \mathrm{CFF}_{1000} + \mathrm{CFF}_L$). Compute the effect of the Landers earthquake in terms of advancing or retarding the time until the next earth-

quake for each fault orientation. Express your answer in years, using the sign convention of positive time for advancement of the next earthquake and negative time for retardation. (Warning: This is tricky.) Check your answer against the values of shear stress on the fault. Generally (but not always) the earthquake time should advance when the long-term and Landers shear changes agree in sign (either both positive or both negative), and the time should be delayed when the shear stress changes disagree in sign.

(i) No increase in seismicity (small earthquake activity) has been observed near PFO following the Landers event. Does this say anything about the validity of the threshold CFF model?

Hint: Getting the signs correct in parts (f)–(h) can be complicated, particularly for part (h). Stresses can be either positive or negative. To help get it right, define two unit vectors for each fault azimuth, one parallel to the fault ($\hat{\mathbf{f}}$) and one perpendicular to the fault ($\hat{\mathbf{p}}$). Compute the traction vector by multiplying the stress tensor by $\hat{\mathbf{p}}$. Then resolve the traction vector into shear stress and normal stress by computing the dot product with $\hat{\mathbf{f}}$ and $\hat{\mathbf{p}}$, respectively. Naturally, $\hat{\mathbf{f}}$ and $\hat{\mathbf{p}}$ must be of unit length for this to work.